MILITARY SPECIFICATION

CABLES, (POWER AND SPECIAL PURPOSE) AND WIRE, ELECTRICAL (300 AND 600 VOLTS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers single and multiple conductor cables with and without shields for use in 300 and 600 volts root mean square (rms) applications (see 6.1).

1.2 Classification. Cables covered by this specification shall be furnished in accordance with the type designation specified and shall be classified by the elements thereof.

1.2.1 Type designation. The type designation shall be formed as follows:

CO-	08	HG	E	(2/16S-4/12 -2/8)	SJ	1090
Component (1.2.1.1)	Total number of conductors (1.2.1.2)	Class (1.2.1.3)	Flexibility (1.2.1.4)	Conductor data (1.2.1.5)	Shielding under jacket (1.2.1.6)	Outside diameter (1.2.1.7)

1.2.1.1 Component. Cables covered by this specification are identified by the two-letter symbol CO followed by a hyphen.

1.2.1.2 Total number of conductors. The total number of conductors (see 6.4.1), in a cable exclusive of uninsulated ground wires (see 6.4.2 and 6.4.5), are identified by a two-digit number. For cables with less than 10 conductors, a "0" preceeds the number of conductors (example - 04 = 4 conductors), in a multiple conductor power cable.

1.2.1.3 Class. The class of cables is identified by a two-letter symbol, the first letter of which designates the duty (see 6.1) in accordance with table I. The second letter indicates the application of the cables in accordance with table II.

1.2.1.4 Flexibility. Flexibility of a cable is based upon the flexibility of the conductors comprising the cable. All conductors and uninsulated ground wires within the same cable shall be of the flexibility designated (see 3.2.1 and 3.2.2). The flexibility is identified as follows:

S - Semiflexible.
F - Flexible.
E - Extra-flexible.
M - Most-flexible.

1.2.1.5 Conductor data.

Beneficial comments (recommendations and additions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Electronics Research and Development Command, ATTN: DELET-R-S, Ft. Monmouth, N.J. 07703 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:40Z
Check the source to verify that this is the current version before use.

1.2.1.5.1 Number and size of individual conductors. The number of individual conductors of the same wire sizes shall be designated by that number followed by a slant line and followed by a number indicating the conductor wire size; for example, "3/22" indicates 3 No. 22 AWG conductors. When cables are made up of different wire sizes, each different wire size shall be individually represented, with a dash separating each different wire size; for example, "-6/14-1/8" indicates six No. 14 AWG conductors and a single No. 8 AWG conductor.

TABLE I. Duty of cables.

(Class) Designating letter	Duty	Voltage rating, (max)	Insulation thickness (nom range), incl.		Jacket thickness (nom range), incl	
			Vulcanized rubber	Polyethylene and PVC	Vulcanized rubber	Polyurethane and PVC
		Volts	Inch	Inch	Inch	
L	Light	300	0.016 to 0.021	0.010 to 0.015	0.020 to 0.050	10% - 15% of jacket inside diameter
M	Medium	600	.030 to .038	.012 to .020	.027 to .172	
H	Heavy	600	.030 to .094	.015 to .075	.035 to .203	

TABLE II. Application of cable.

(Class) Designating letter	Application	Insulating compound	Jacketing compound
		Per MIL-I-3930	
G	General purpose, medium low temperature, -40° to +75°C	IS, IH, 2/ SBR, 3/ EPM or EPDM	JS, JN (SBR)
O	Oil resistant, medium low temperature, -40° to +75°C	IS, IP (SBR)	JN Chloroprene rubber
L	Low temperature, heat resistant, -55° to +75°C	IS-L (SBR)	JS-L, JN-L (SBR)
D	Low temperature, oil resistant, heat resistant, -55° to +75°C	IS-L (SBR)	JN-L Chloroprene rubber
H	Low temperature, superior heat and weather resistant, -55° to +90°C	IH (EPM or EPDM)	JH (EPDM)
B	Low temperature, oil resistant, superior heat resistant, -55° to +80°C	IH (EPM or EPDM)	JN-L Chloroprene rubber
C	Low temperature, oil resistant, heat resistant, -55° to +80°C	IE Polyethylene crosslinked	JN-L Chloroprene rubber
K	Low temperature, oil resistant, heat resistant, weather resistant, -55° to +80°C	IE	JU Thermoplastic polyurethane elastomer
N	Low temperature, oil resistant, heat resistant, -40° to +85°C 1/	IR or IL	JR or JN Chloroprene rubber

1/ Synthetic polyisoprene may be blended with natural rubber.
2/ Styrene - Butadiene rubber.
3/ Ethylene - Propylene rubber or ethylene-propylene-diene rubber.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:40Z
Check the source to verify that this is the current version before use.

1.2.1.5.2 Unshielded grouped conductors. Conductors twisted together to form a group (such as pairs, triples, quads, etc.) are designated, immediately following the wire size, by "x", followed by the number of these conductor groups contained in the cable; for example, "2/20 x 1" indicates two No. 20 AWG conductors twisted together to form a pair; "3/22 x 4" indicates four groups, each group consisting of three No. 22 AWG conductors twisted together to form a triple.

1.2.1.5.3 Shielding.

1.2.1.5.3.1 Grouped shielding. When one or more identical groups of copper alloy conductors have a common shield over each conductor group, the designation "Sx (No. of groups)" follows the wire size; for example, "2/22 S x 6" indicates six pairs of No. 22 AWG conductors, with a shield over each pair. When a single group has an overall shield, the designation "x1" is to be omitted since conductors under a shield are twisted to form one group; for example, "2/20 S" indicates one twisted pair of No. 20 AWG conductors under a common shield.

1.2.1.5.3.2 Individual shielding. When each conductor of the same size is individually shielded, the letter "I" follows the letter "S"; for example, "3/14SI" indicates three No. 14 AWG conductors, each conductor individually shielded. A shielded single conductor is to be designated by the letter "S" instead of "SI"; for example "1/18 S" indicates a single No. 18 AWG conductor shielded.

1.2.1.5.4 Uninsulated ground wires. Uninsulated ground wires are designated in the same manner as conductors specified in 1.2.1.5 except the letter "R" is to follow the wire size; for example, "3/8 R" indicates three No. 8 AWG ground wires. Uninsulated ground wires are not counted in the total number of conductors.

1.2.1.5.5 Cabling. All conductors and groups that are cabled together beneath the jacket, and under the jacket shield, if present, shall be enclosed in parentheses; for example, "(2/20 x 2-3/18)" indicates a cable made up of two twisted pairs No. 20 AWG conductors and three No. 18 AWG conductors.

1.2.1.5.6 Cable configuration. Cables composed of two conductors laid parallel are to be designated by the letter "F" which is to follow the conductor number; for example, "2F/8" indicates two No. 8 AWG conductors laid parallel.

1.2.1.6 Shielding under jacket. When shielding is over all conductors and beneath the jacket, the letters "SJ" follows the parentheses that inclose the conductor data. No letters are used if an overall shield beneath the jacket is not present.

1.2.1.7 Outside diameter.

1.2.1.7.1 Cables (except parallel conductors). The outside diameter of the cable is indicated by four significant figures expressing the diameter in mils (thousandths of an inch) to the nearest figure divisible by five. If a tolerance other than that specified in 3.2.8.1 is required, it is indicated by placing the tolerance immediately after this element of the type designation (see 1.2.1.8b and 1.2.1.8f). Special tolerance cables are inactive for new design.

1.2.1.7.2 Parallel conductor cables. The outside diameter of the parallel conductor cable shall be designated by giving the major diameter followed by a slant and followed by the minor diameter. The diameter, either major or minor, shall be indicated in mils (thousandths of an inch) by four significant figures. If a tolerance other than that specified 3.2.8.2 is required, it shall be indicated by placing the tolerance immediately after the major or minor diameter. Special tolerance cables are inactive for new design.

1.2.1.8 Use and examples of type designations. Use of type designations formed as indicated in 1.2.1 is not intended to limit the number of cable types procured under this specification. Representative examples of complete type designations are as follows:

a. CO-10 LGF (6/22 - 2/20 - 2/18 x 1)0640 - Represents a 10-conductor light duty, general purpose flexible cable with six no. 22 AWG conductors, two No. 20 AWG conductors and one twisted pair No. 18 AWG conductors. The cable has an outside diameter of 0.640 inch.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:40Z
Check the source to verify that this is the current version before use.

b. CO-18 HLF (3/20 x 4-3/18 x 1 - 3/16)1040 ±0025 - Represents an 18-conductor, heavy duty, heat resistant flexible cable with four triples, each triple composed of three No. 20 AWG conductors twisted together; one triple composed of three No. 18 AWG conductors twisted together, and three No. 16 AWG conductors. The cable has an outside diameter of 1.040 inches with a tolerance ±0.025 inch specified.
c. CO-06 MOE (3/22 S - 2/20 SI - 1/16 S)SJ0430 - Represents a 6-conductor, medium duty, oil resistant extra flexible cable with three No. 22 AWG conductors twisted together to form a triple, in a common shield; two No. 20 AWG conductors, each individually shielded and one 16 AWG conductor individually shielded. The cable has an overall shield with a jacket or sheath. The cable has an outside diameter of 0.430 inch.
d. CO-15 MDF (2/22 SX5 - 2/20S-2/18SI-1/16S)1620 - Represents a 15 conductor, medium duty, oil and heat resistant flexible cable with five pairs of No. 22 AWG conductors, each pair individually shielded; one shielded pair of No. 20 AWG conductors; two No. 18 AWG conductors, each individually shielded; one No. 16 AWG shielded conductor. The cable has an outside diameter of 1.620 inches.
e. CO-02 HGF (2/4 - 2/8R)1080 - Represents a 2-conductor, heavy duty, general purpose flexible cable with two No. 4 AWG conductors, and two No. 8 AWG uninsulated ground wires. The cable has an outside diameter of 1.080 inches.
f. CO-02 HGS (2F/8)0920 ±0050/0510 - Represents a 2-conductor, heavy duty, general purpose, semi-flexible parallel cable with two No. 8 AWG conductors having a major diameter of 0.920 inch with a tolerance of ±0.050 inch specified and a minor diameter of 0.510 inch.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-P-390	-	Plastic, Molding and Extrusion Material, Polyethylene and Copolymers (Low, Medium, and High Density).
Q-C-265	-	Chemical, Analytical, General Specification For.
QQ-W-343	-	Wire, Electrical (Uninsulated).
TT-I-735	-	Isopropyl Alcohol.
TT-S-735	-	Standard Test Fluids, Hydrocarbon.

MILITARY

MIL-C-572	-	Cords, Yarns and Monofilaments, Organic Synthetic Fiber.
MIL-I-631	-	Insulation, Electrical, Synthetic-Resin Composition, Nonrigid.
MIL-I-3930	-	Insulating and Jacketing Compounds, Electrical (For Cable, Cords, and Wires).
MIL-L-6082	-	Lubricating Oil, Aircraft Reciprocating Engine (Piston).
MIL-C-12000	-	Cable, Cord, and Wire, Electric; Packaging of.
MIL-M-20693	-	Molding Plastic, Polyamide (Nylon), Rigid.
MIL-P-24216	-	Polypropylene Cores, Strand Centers and Substrands for Wire Rope.

STANDARDS

FEDERAL

FED-STD-228	-	Federal Test Method Standard Cable and Wire, Insulated; Methods of Testing.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:40Z
Check the source to verify that this is the current version before use.

MILITARY

MIL-STD-104 - Limits for Electrical Insulation Color.
MIL-STD-105 - Sampling Procedures and Tables For Inspection by Attributes.
MIL-STD-686 - Cable and Cord Electrical; Identification Marking and Color Coding of.
MIL-STD-45662 - Calibration Systems Requirements.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ANSI/ASTM D 3699-78 Kerosene.

(Copies of ASTM publications may be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Material. Material shall be as specified herein. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the product. When a definite material is not specified, a material shall be used which will enable the wire and cable to meet the requirements of this specification.

3.1.1 Wire strands.

3.1.1.1 Copper strands. The strands comprising the conductors, uninsulated ground wires (see 6.4.5), and braided shields shall be coated, soft or drawn, and annealed wire conforming to the applicable requirements of QQ-W-343.

3.1.2 Insulating and jacketing compounds. The insulating and jacketing compounds shall comply with the applicable types (see table II). Polyamide shall be in accordance with MIL-M-20693, composition A, type III, grade E.

3.1.3 Textiles. Textiles used for separators, fillers, and binders shall consist of synthetic thread or yarn, as applicable, conforming to MIL-C-572, twisted polyethylene film or natural jute material. Foam polypropylene is acceptable for use with thermoplastic materials only. The yarn shall be clean, dry, and substantially free from all foreign particles, knots, lumps, or any substance that would impair the insulation of the conductor.

3.1.4 Polyester tape. Polyester tape shall conform to MIL-I-631, type G, form T_F. Polyester tape may be polyethylene-coated for subsequent heat sealing.

3.1.5 Polypropylene tape. Polypropylene tape shall be in accordance with MIL-P-24216.

3.1.6 Polyethylene. Polyethylene shall be in accordance with L-P-390.

3.1.7 Recycled material. These cables and wires shall contain recycled material to the maximum extent possible provided such material does not prevent the manufacturer from meeting the requirements of this specification. The recycled material shall be reprocessed, remanufactured in a manner which will restore it to the same chemical composition and physical properties as the material originally selected.

3.1.8 Polyurethane. Polyurethane shall be of the polyetherurethane type.

3.2 Design and construction. Cable shall be of the design, construction, and physical dimensions specified herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:40Z
Check the source to verify that this is the current version before use.

3.2.1 Conductors. Conductors shall be in accordance with QQ-W-343. For a particular size and flexibility, the conductor shall be of the type, class, and service listed in table III (see 1.2.1.4).

3.2.2 Uninsulated ground wires (power cables only). Uninsulated ground wires shall be in accordance with QQ-W-343 and table IV. Each uninsulated ground wire shall have a green cover of braid, rubber, or plastic (see 3.2.6.2).

3.2.3 Insulated conductors. The insulation shall be free-stripping and extruded over the conductor. Insulated conductors shall be readily separable from each other. The minimum thickness, measured at any cross section of the insulation, for the various cable duties and conductor sizes, shall be as specified in table V. The minimum insulation thickness; however, may be 80 percent of the nominal thickness at the area of contact between contiguous insulated conductors. There shall be no kinking of the insulated conductors within the sheath. Polyethylene insulated conductors of 24 through 12 AWG shall have 0.003-inch minimum thickness extruded jackets of polyamide.

3.2.4 Shielding. Shielding shall be applied over an individual conductor, over two or more conductors, or over the entire group of conductors, as specified (see 1.2.1.5.3). The shield shall consist of a braid providing a minimum coverage of 85 percent (see 4.6.1). Irregularities, breaks, and discontinuities shall be avoided in the application. There shall be no strands protruding through the insulation or the outer jacket. For insulated conductors in MCM and AWG sizes 500 MCM to 10 AWG, inclusive, the shielding shall consist of a single braid of strands, each strand having a nominal diameter of 0.0063 inch; for insulated conductors in AWG sizes No. 12 to 22, inclusive, the strands shall be of 0.0050 inch nominal diameter. For a shield applied over two or more conductors grouped together, or over the entire group of conductors, when the diameter covered is 0.625 inch or less, the shielding shall consist of a single braid of strands, each with a nominal diameter of 0.0050 or 0.0063 inch; when the diameter covered is more than 0.625 inch, strands with a nominal diameter greater than 0.0063 inch may be used for the braid.

3.2.4.1 Shield cover. For a single conductor, or for two or more conductors grouped together under one shield, there shall be a polyester tape overlapping-helical wrap applied directly over the shield (extruded nylon permitted in lieu of tape). When the diameter over the shield does not exceed 0.20-inch, 0.005 to 0.008-inch thick polyamide may be extruded instead of polyester tape over shields that have polyethylene insulated components. For an overall shield, there shall be an overlapping-helical wrap of polyester tape applied directly over the cable core prior to shielding.

3.2.5 Identification coding. The individual conductors of all cables and single wires shall be coded for their entire length in accordance with MIL-STD-686.

a. The standard means of coding shall be as follows:
   1. Solid-colored insulation, with colored-stripe tracer, when required.
   2. Color-coded 1/2 to 1 mil polyester tape, with or without 1/2 to 1 mil heat-sealable adhesive.
   3. Color-coded 1 mil polyester tape with 1/2 to 1 mil heat-sealable adhesive in 8 AWG or larger, when tracers are required.
   4. Color-coded, filled tapes in 8 AWG and larger, when tracers are not required.
   5. Coding by marking.
   6. Color code by green braid (ground wires only).

b. Color shall be in accordance with MIL-STD-104. Class 1 for plastics and elastomers, class 2 for thermosetting compounds. Where black, blue, brown, purple, or red (i.e. two or more of these colors) are used in the same cable, the colors (other than black) shall be light to nominal. Gray shall be dark (or slate) when in the same cable with white.

3.2.5.1 Solid-colored insulation. The entire thickness of rubber insulation shall be solid-colored, when practicable. When the entire thickness is not solid-colored, a thin layer of solid-colored insulation shall be applied over the inner core to form an integral part of the insulation wall. Cables having up to and including six conductors shall employ solid coloring, and each conductor shall be clearly distinguishable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:40Z
Check the source to verify that this is the current version before use.

TABLE III. Constructional and direct-current resistance requirements of conductors.

Conductor size MCM and AWG	Conductor dc resistance at 25°C max ohms/1000 ft	Semiflexible		Number of strands	Flexible			Number of strands	Extra flexible			Number of strands	Most flexible		Number of strands
		Type	Class		Type	Class	Service		Type	Class	Service		Type	Class	
		QQ-W-343			QQ-W-343				QQ-W-343				QQ-W-343		
500	0.023	E	G	259	E	H	-	427	-	-	-	-			
450	.025	E	G	259	E	H	-	427	-	-	-	-			
400	.028	E	G	259	E	H	-	427	-	-	-	-			
350	.032	E	G	259	E	H	-	427	-	-	-	-			
300	.038	E	G	259	E	H	-	427	-	-	-	-			
250	.045	E	G	259	E	H	-	427	-	-	-	-			
4/0	.054	E	G	133	E	H	-	259	-	-	-	-	R	M	5320
3/0	.069	E	G	133	E	H	-	259	-	-	-	-	R	M	4256
2/0	.086	E	G	133	E	H	-	259	-	-	-	-	R	M	3323
1/0	.113	E	G	133	E	H	-	259	R	K	-	1064	R	M	2646
1	.142	E	G	133	E	H	-	259	R	K	-	836	R	M	2107
2	.180	E	G	49	E	1/	-	259	R	K	-	665	R	M	1666
3	.213	E	G	49	E	H	-	133	R	K	-	532	R	M	1323
4	.283	E	G	49	E	H	-	133	R	K	-	420	R	M	1064
6	.450	E	G	49	E	H	-	133	R	K	-	266	R	M	665
8	.715	--	-	-	E	G	-	49	R or E	K H	-	168 133	R	M	420
10	1.14	--	-	-	B	K	Severe	104	R	M	-	259			
12	1.81	--	-	-	B	K	Severe	65	B or R	M M	Extra severe -	165 168			
14	2.82	--	-	-	B	K	Severe	41	B	M	Extra severe	104			
16	4.49	B	K	26	B	M	Severe	65	B	O	Extra severe	104			
18	7.15	B	K	16	B	M	Severe	41	B	O	Extra severe	65			
20	11.35	--	-	-	B	M	Severe	26	B	O	Extra severe	41			
22	18.09	--	-	-	B	M	Severe	19	B	O	Extra severe	26			
24	24.50	--	-	-	B	O	Severe	19	B	P	Extra severe	26			

1/ Number 2 AWG conductor shall be type RC, except it shall be composed of 259-16.0 mil strands.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:40Z
Check the source to verify that this is the current version before use.

TABLE IV. Constructional and direct-current-resistance requirements of uninsulated ground wires.

Ground wire size	Ground wire resistance at 25°C max	Semiflexible				Flexible				Extra flexible				Length of lay max 1/
		No. of strands	Dia. of strands	Type	Class	No. of strands	Dia. of strands	Type	Class	No. of strands	Dia. of strands	Type	Class	
AWG	Ohms/1000 ft		Mils	QQ-W-343			Mils	QQ-W-343			Mils	QQ-W-343		Inches
3/0	.069	133	35.5	E	G	259	25.5	E	H	---	---	--	---	---
2/0	.086	133	31.6	E	G	259	22.7	E	H	---	---	--	---	---
1/0	.113	133	28.2	E	G	259	20.2	E	H	---	---	--	---	---
1	.142	133	25.1	E	G	259	18.0	E	H	---	---	--	---	---
2	.173	49	36.8	E	G	133	22.3	E	H	---	---	--	---	---
3	.218	49	32.8	E	G	133	19.9	E	H	---	---	--	---	---
4	.283	49	29.2	E	G	133	17.7	E	H	420	10.0	R	K	---
5	.350	49	26.0	E	G	133	15.8	E	H	336	10.0	R	K	---
6	.450	49	23.1	E	G	133	14.0	E	H	266	10.0	R	K	---
7	.556	19	33.1	C	C	49	20.6	E	G	210	10.0	R	K	---
8	.715	19	29.5	C	C	49	18.4	E	G	168	10.0	R	K	---
9	.884	19	26.2	C	C	49	16.4	E	G	133	10.0	R	K	---
10	1.11	7	38.5	C	B	19	23.4	C	C	104	10.0	B	K	2.00
11	1.41	7	34.3	C	B 2/	19	20.8	C	C 2/	82	10.0	B	K 2/	1.50
12	1.77	7	30.5	C	B	19	18.5	C	C	65	10.0	B	K	1.25

1/ Since many of these sizes are not listed in QQ-W-343, the maximum length of lay is specified herein, instead of the service. This is the equivalent of extra severe service.
2/ Since these sizes are not listed in QQ-W-343, the applicable class is listed for information only.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:40Z
Check the source to verify that this is the current version before use.

TABLE V. Insulation thickness.

Duty (see table I)	Rubber			Polyethylene		
	Conductor size, inclusive AWG and MCM	Insulation thickness		Conductor size, inclusive AWG and MCM	Insulation thickness	
		Nominal 1/	Minimum		Nominal 1/	Minimum
		inch	inch		inch	inch
Light	24 to 18	0.016	0.013	24 to 18	0.010	0.008
	16 to 14	.018	.015	16 to 14	.012	.010
	12 to 10	.021	.018	12 to 10	.015	.012
Medium	24 to 20	0.030	0.025	24 to 18	0.012	0.010
	18 to 16	.033	.028	18 to 16	.014	.012
	14 to 12	.036	.031	14 to 12	.016	.014
	10 to 8	.038	.033	10 to 8	.020	.018
Heavy	24 to 20	0.030	0.025	24 to 16	0.015	0.012
	18 to 16	.033	.028	14 to 12	.020	.017
	14 to 10	.047	.042	10	.030	.025
	8 to 6	.055	.050	8 to 4	.035	.032
	4 to 2	.063	.057	2	.042	.038
	1	.070	.063	1	.046	.041
	0 to 4/0	.078	.070	1/0 to 2/0	.056	.052
	200 to 500 MCM	.094	.084	3/0 to 4/0	.064	.060
				200 to 500 MCM	.075	.070

1/ Nominal dimensions are included for information only.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:40Z
Check the source to verify that this is the current version before use.

3.2.5.2 Stripe tracers. The tracers shall be in accordance with MIL-STD-686. All materials for striping shall be nonconductive.

3.2.5.3 Coding by marking. Where prototype models or short runs of cables are specified for procurement (less than 10,000 feet) and the individual conductors of the cables have an outside diameter of 0.065 inch or larger, the contractor shall code the individual conductors by marking or coloring in accordance with MIL-STD-686.

3.2.6 Grouping and cabling. When cables are grouped (see 1.2.1.5.2 and 1.2.1.5.5), the conductors within the group shall be twisted together. All conductors and groups of conductors, shall be cabled together in an arrangement that shall form an essentially round construction (except for parallel conductors). Where there are mixtures of different conductor sizes and element constructions (shielded singles, shielded pairs), the larger diameter conductors and shielded wires shall be toward the center.

3.2.6.1 Length of lay. The lay-up of cable layers shall be unidirectional. For cables employing semiflexible conductors and uninsulated ground wires, when present, the length of lay shall be not greater than 20 times the outer diameter of the layer. For cables employing flexible conductors and uninsulated ground wires, when present, the length of lay shall be not greater than 16 times the outer diameter of the layer. For cables employing extra-flexible conductors and uninsulated ground wires, when present, the length of lay shall be not greater than 12 times the outer diameter of the layer. (The outer diameter of the layer is the diameter over the separator or binder, if used (see 3.2.6.4 and 6.1.6).

3.2.6.2 Uninsulated ground wires. When specified (see 1.2.1.5.4), uninsulated ground wires shall be included in two, three, four and five conductor, heavy duty unshielded cables in which all conductors are of the same size, and shall be cabled together with the conductors and fillers, if used. The number of uninsulated ground wires shall be the same as the number of conductors. The minimum size of the uninsulated ground wires shall be as specified in Table VI and is dependent upon the size and number of conductors. Uninsulated ground wires shall not be furnished in conductor sizes less than 8 AWG.

TABLE VI. Size of uninsulated ground wires.

Power conductor size MCM and AWG	Number of power conductors		
	Two	Three	Four
	Minimum uninsulated ground wire size, AWG		
500	4/0	2/0	1/.0
400	3/0	1/0	1
350	2/0	1	2
250	1/0	2	3
4/0	1	3	4
3/0	2	4	5
2/0	3	5	6
1/0	4	6	7
1	5	7	8
2	6	8	9
3	7	9	9
4	8	10	11
6	10	10	12
8	10	12	12

3.2.6.3 Fillers. Fillers of jute or synthetic textiles shall be used when required to effect a circular cross section (see 3.1.3).

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:40Z
Check the source to verify that this is the current version before use.

3.2.6.4 Binders and separators. A binder applied over the cabled conductors to hold them in place is optional. A separator shall be applied over the cabled conductors prior to the application of the shield or jacket. The use of a separator for unshielded conductors of 8 AWG and larger shall be optional with the manufacturer. A separator shall be applied over the shield, if present, prior to the application of the jacket. For single conductor cables where no shield is present, a separator shall not be applied. A core wrap shall be applied as a heat barrier over cabled conductors with polyethylene insulation, prior to extrusion of the jacket. The binders and separators shall be of the types specified in table VII.

TABLE VII. Binders and separators.

Binder types (optional)	Separator types		
	For unshielded cable	For shielded cable	
		Under shield	Over shield
Open wrap, open braid, or polyester tape.	Closed wrap, closed braid, compound-filled tape, polyester or polypropylene tape.	Polyester or polypropylene tape.	Closed wrap, closed braid, compound-filled tape, polyester or polypropylene tape.

3.2.7 Jacket. The jacket shall be extruded and then vulcanized (except for thermoplastics and thermoplastic elastomers) over the cabled conductors or shielding, when used, and shall be well centered, single- or double- layer jacket. Double-layer jackets shall be used on cables whose minimum jacket thickness is 0.098 inch minimum or over as specified in table VIII. Double-layer jackets shall be applied in two concentric layers so vulcanized or cured that they are strongly bonded together; the outer layer shall be at least 50 percent of the total thickness. A reinforcement consisting of an open braid or two layers of seine twine or cabled cotton, or the equivalent, applied in reverse directions, shall be provided between the layers of the jacket (see 6.1.7). For single conductor cables where no shield is present, the jacket shall be applied directly over the insulation and adhere to the insulation. The jacket thickness shall be as specified in table VIII. The jacket surface shall not contain ridges or seams, and shall present a smooth appearance without any objectionable roughness or irregularities, consistent with good industry practice.

3.2.7.1 Polyurethane jacket. Thickness for polyurethane jacket shall be 10-15 percent of the cable core diameter (jacket inside diameter). For medium and heavy duty cable, the jacket shall be reinforced with polyester or glass fiber strands treated, (if necessary) to bond to polyurethane. The fibers shall be applied over the cable core and bonded to the jacket.

TABLE VIII. Jacket thickness.

Calculated core diameter, inches, inclusive	Jacket thickness					
	Duty L		Duty M		Duty H	
	Nominal	Minimum	Nominal	Minimum	Nominal	Minimum
0.125 and under	0.020	0.018	0.027	0.024	0.035	0.032
.126 to 0.155	.022	.019	.031	.028	.040	.036
.156 to .219	.024	.022	.039	.035	.045	.041
.220 to .234	.026	.023	.039	.035	.078	.070
.235 to .290	.031	.028	.047	.042	.078	.070
.291 to .300	.031	.028	.047	.042	.094	.085
.301 to .430	.050	.045	.063	.057	.094	.085
.431 to .540	---	---	.070	.063	.094	.085
.541 to .640	---	---	.078	.070	.109	.098
.641 to .740	---	---	.094	.085	.125	.113
.741 to .850	---	---	.109	.098	.141	.127
.851 to 1.1000	---	---	.125	.113	.156	.140
1.101 to 1.320	---	---	.156	.140	.172	.155
1.321 to 1.550	---	---	.172	.155	.188	.169
1.551 to 1.820	---	---	---	---	.203	.183

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:40Z
Check the source to verify that this is the current version before use.

3.2.8 Diameter tolerance.

3.2.8.1 Cable diameter tolerance (except parallel conductor). The outside diameter of cables having a specified diameter of less than 0.500 inch shall have a tolerance of ±0.015 inch. The outside diameter for cables having a specified diameter from 0.500 inch to 0.700 inch, inclusive shall have a tolerance of ±0.020 inch. The outside diameter for cables having a specified diameter of more than 0.700 inch shall have a tolerance of ±3 percent. The diameter of a cable at any cross section (except for parallel conductor cables) shall be considered the average of the major and minor diameters at that cross section. If a special tolerance is required on the outside diameter, such tolerances shall be indicated at the end of the type designation (see 1.2.1.8).

3.2.8.2 Parallel conductor. The outside diameter of parallel conductor cables shall have a tolerance of ±0.040 inch for the major diameter and a tolerance of ±0.030 inch for the minor diameter.

3.3 Spark (optional for polyethylene). When the cable is tested as specified in 4.6.2, the insulated conductor shall withstand the applicable voltage specified in table XIII. A certification to the effect that the spark test has been performed shall accompany the cable when it is submitted for acceptance.

3.4 Voltage breakdown. When tested as specified in 4.6.3 or 4.6.4, the insulated conductor in the finished cable shall withstand the specified voltage in table XIII. Polyethylene insulation shall be impulse tested.

3.5 Insulation resistance. When measured as specified in 4.6.5, the insulation resistance shall be not less than the figure obtained from the following formula:

$$R = K \log_{10} \frac{d+2c}{d}$$

Where:
R = Insulation resistance in megohms -1000 feet.
K = 2,000 for SBR, EPDM, and PVC; 50,000 for polyethylene.
d = Average diameter under the insulation.
c = Specified nominal insulation thickness.
d and c shall be expressed in the same units.

3.6 Direct-current resistance. When the cable is tested as specified in 4.6.6, it shall not exceed the values shown in tables III and IV.

3.7 Surface resistance. Black jackets, black insulation, and black striped jackets and insulation shall have a surface resistance of 10 megohm-inches minimum when tested as specified in 4.6.7.

3.8 Cold bend. When the cable is tested as specified in 4.6.8, the insulation, jacket, and inked markings shall not crack.

3.9 Bend. When finished cable, except parallel conductor construction, is tested in accordance with 4.6.9, there shall be no electrical breakdowns, cracking, or splitting of the outer sheath during or at the completion of the test. (Specimens used for this test shall not be used for additional testing).

3.10 Durability (polyethylene and polyamide insulation jackets only). When the cable is tested as specified in 4.6.10, there shall not be a continuous line evident through any letter or numeral of the ink marking nor through any striping applied to the outer surface of polyethylene or polyamide material when subjected to 250 cycles (500 strokes) of abrasive action.

3.11 Polyethylene shrink-back. When the cable is tested as specified in 4.6.11, it shall shrink back a maximum of 1/16 inch and shall not split or crack.

3.12 Polyamide heat stability. When the cable is tested as specified in 4.6.12, there shall be no tears or cracks in the polyamide materials. Wrinkles or softening and flowing of the polyamide shall not be cause for rejection.

3.13 Fluid immersion (class HN only). When tested as specified in 4.6.13, the increase in the diameter of the cable shall be less than 15 percent, the cable shall then meet the voltage breakdown requirements of 3.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:40Z
Check the source to verify that this is the current version before use.

3.14 Marking. The cable jacket or marker tape shall be marked in accordance with MIL-STD-686, in addition, the manufacturer's name or trademark and the year of manufacture shall be included. The outer surface of the cables shall be smooth and free from raised or indented markings. All letters and numbers in the marking shall be of the same height. There shall be no spacing between letters and numbers in the type designation, for example, CO-06HOF(6/18)0524.

3.15 Workmanship. Cables shall be constructed and finished in a thoroughly workmanlike manner in accordance with accepted high grade production techniques. The cables shall be a uniform and consistent product and shall be free from any defects which will adversely affect the serviceability of the product, such as lumps, kinks, splits, abrasions, scrapes, corroded surfaces, skin impurities and faulty extruded surfaces.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Test equipment and inspection facilities. Test and measuring equipment and inspection facilities of sufficient accuracy, quality and quantity to permit performance of the required inspection shall be established and maintained by the contractor. The establishment and maintenance of a calibration system to control the accuracy of the measuring and test equipment shall be in accordance with MIL-STD-45662.

4.2 Classification of inspections. The inspections specified herein are classified as follows:

a. Materials inspection (see 4.3).
b. Quality conformance inspections (see 4.5).

4.3 Materials inspection. Materials inspection shall consist of certification supported by verifying data that the materials listed in table X used in fabricating the cables, are in accordance with the applicable referenced specifications or requirements prior to such fabrication.

TABLE IX. Materials inspection.

Material	Requirement paragraph	Applicable specification
Copper	3.1.1.1	QQ-W-343
Insulating and jacketing compounds	3.1.2	MIL-I-3930 or MIL-M-20693
Textiles	3.1.3	MIL-C-572
Polyester tape	3.1.4	MIL-I-631
Polypropylene tape	3.1.5	MIL-P-24216
Polyethylene	3.1.6	L-P-390

4.3.1 Reinspection. If the cable is supplied on spools, reels, or coils, or in precut lengths and is more than 12 months or 4 quarters of a year old from date of last inspection to date of shipment, the cable shall be reinspected to meet the applicable aging test of MIL-I-3930, or MIL-M-20693 for the insulating and jacketing compounds used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:40Z
Check the source to verify that this is the current version before use.

4.4 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in the "GENERAL REQUIREMENTS" of FED-STD-228.

4.5 Quality conformance inspection.

4.5.1 Inspection of product for delivery. Inspection of product for delivery shall consist of groups A, B and C inspection.

4.5.1.1 Inspection lot. An inspection lot shall consist of all cables of the same type designation (see 1.2.1) produced under essentially the same conditions, and offered for inspection at one time.

4.5.1.2 Unit of product. The unit of product shall be taken as a continuous length of cable contained on a spool, reel, or coil. The unit of product shall not exceed 5,000 feet of cable on each spool, reel, or coil.

4.5.1.3 Sample. The sample shall consist of that number of randomly selected units of product required by the applicable sampling plan for the presented lot.

4.5.1.4 Selection of sample units. Sample units for inspection shall be taken from each unit of product which forms a part of the sample. A sample unit is defined as a length of cable drawn from a unit of product.

4.5.1.5 Test specimen. A test specimen may be the entire sample unit (length of finished cable) or any portion of the sample unit that is to be tested.

4.5.1.6 Group A inspection. Group A inspection shall consist of the inspections specified in table X, in the order shown.

4.5.1.6.1 Sampling plan. Statistical sampling and inspection shall be in accordance with MIL-STD-105 for general inspection level II. The acceptable quality level (AQL) shall be as specified in table X.

4.5.1.6.2 Major and minor defects. Major and minor defects shall be as defined in MIL-STD-105 and as classified below. Any sample unit which has one or more major or minor defects shall be a defective.

a. Major defect:
   - Shrink-back of polyethylene insulation.
   - Wrong color code.
   - All electrical defects.
   - Broken strands in conductor.
   - Insulation thickness.
   - Nicked or torn insulation.
   - Occlusions in insulation.
   - Jacket thickness not within requirements.
   - Outside diameter incorrect.
   - Poor conductor coating.
   - Incomplete coverage of separator.
   - Incorrect concentricity of insulation.
   - Fails stripping durability (polyamide and polyethylene only).

b. Minor defect:
   - Missing strands in conductor.
   - Poor stranding tension.
   - Poor free stripping.
   - Rough surface of insulation.
   - Color range of insulation incorrect.
   - Wrong footage.
   - Stripe width and lay incorrect.

4.5.1.6.3 Rejected lots. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:40Z
Check the source to verify that this is the current version before use.

TABLE X. Group A inspection.

Examination	Requirement paragraph	Test method paragraph	AQL (percent defective) Major	AQL (percent defective) Minor
Visual and dimensional				
Conductors	3.2.1	4.5.1	1.0	4.0
Uninsulated ground wires	3.2.2			
Insulated conductors	3.2.3			
Shielding	3.2.4			
Identification coding	3.2.5			
Grouping and cabling	3.2.6			
Jacket	3.2.7			
Diameter tolerance	3.2.8			
Marking	3.14			
Workmanship	3.15			
Electrical (in order shown)				
Spark test	3.3	4.6.2	100	
Voltage breakdown	3.4	4.6.3 or 4.6.4		
Insulation resistance	3.5	4.6.5		
Direct current resistance (conductors and uninsulated ground wires)	3.6	4.6.6	1.0	
Surface resistance	3.7	4.6.7		

4.5.1.7 Group B inspection. Group B inspection shall consist of the inspections specified in table XI and shall be made on sample units which have been subjected to and have passed the group A inspection.

4.5.1.7.1 Sampling plan. The sampling plan shall be in accordance with MIL-STD-105 for special inspection level S-2. The sample size shall be based on the inspection lot size from which the sample was selected for group A inspection. The AQL shall be 6.5 percent defective.

4.5.1.7.2 Rejected lots. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

TABLE XI. Group B inspection.

Inspection	Requirement paragraph	Test method paragraph
Fluid immersion	3.13	4.6.13
Cold bend	3.8	4.6.8
Bend	3.9	4.6.9
Durability	3.10	4.6.10

4.5.1.7.3 Disposition of sample units. Sample units which have passed all the group B inspection shall not be delivered on the contract.

4.5.2 Periodic inspection. Periodic inspection shall consist of group C, except where the results of these inspections show noncompliance with the applicable requirements, delivery of products which have passed group A and B shall not be delayed pending the results of these periodic inspections.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:40Z
Check the source to verify that this is the current version before use.

4.5.2.1 Group C inspection. Group C inspection shall consist of the inspections specified in table XII. Group C inspection shall be made on sample units selected from inspection lots which have passed the groups A and B inspection.

4.5.2.1.1 Sampling plan. One sample shall be selected per each 5,000 feet of cable or fraction thereof, not to exceed a total of five samples per order. Samples shall be selected periodically through the life of the contract to assure representative results.

4.5.2.1.2 Failures. If any sample unit fails to pass group C inspection, the sample shall be considered to have failed.

TABLE XII. Group C inspection.

Inspection	Requirement paragraph	Test method paragraph
Polyethylene shrink-back	3.11	4.6.10
Polyamide heat stability	3.12	4.6.12

4.5.2.1.3 Noncompliance. If a sample fails to pass group C inspection, the manufacturer shall notify the procuring activity and the cognizant inspection activity of such failure and take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which were manufactured under essentially the same materials and processes, and which are considered subject to the same failure. Acceptance and shipment of the product shall be discontinued until corrective action has been taken. After the corrective action has been taken, group C inspection shall be repeated on additional sample units (all inspections, or the inspection which the original sample failed, at the option of the procuring activity). Group A inspection may be reinstituted; however, final acceptance and shipment shall be withheld until the group C inspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure shall be furnished to the cognizant inspection activity.

4.5.3 Inspection of packaging. The sampling and inspection of the preservation, packing, and container marking shall be in accordance with the requirements of MIL-C-12000.

4.6 Methods of inspection.

4.6.1 Visual and mechanical inspection. Cable shall be inspected to verify that the design, construction of conductors, uninsulated ground wires, insulated conductors, shielding, identification coding, grouping and cabling, jacket, diameter tolerances, marking, and workmanship are in accordance with the applicable requirements (see 3.2.1 through 3.2.8, 3.13, and 3.14). The inspections shall be made on a specimen not less than 2 feet in length and cut no closer than 5 feet from the end of the sample.

4.6.2 Spark test (see 3.3). All insulated conductors shall be subjected to the spark test in accordance with method 6211 of FED-STD-228 with the following exceptions:

a. It shall be performed on all insulated conductors prior to cabling, or shielding.
b. For conductors larger than No. 6 AWG, an adaption of method 6211 satisfactory to the Government inspector shall be used.
c. Speed shall be adjusted to insure that contact is maintained for at least .25 seconds between the electrode and any point on the insulation.
d. The frequency of the voltage shall be 60 or 400 Hz.
e. Use of voltmeter-equipped spark testers is optional.

4.6.3 Impulse test (see 3.4). The impulse dielectric test shall be performed in accordance with NEMA Standard No. HP-1-1979 to meet the requirements of 3.4 with the exception that if any dielectric failures occur they must be cut out or suitably identified for subsequent removal along with a minimum of 6 inches of conductor on each side of the fault. Test voltage shall be in accordance with table XIII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:40Z
Check the source to verify that this is the current version before use.

TABLE XIII. Dielectric withstanding voltage.

Dielectric withstanding voltage		Impulse dielectric	
Rubber and all jackets		Polyethylene and all jackets	
Specified nominal insulation thickness, incl. (inch)	Applied DW voltage (volts, rms)	Specified nominal insulation jacket thickness, incl. (inch)	Applied impulse voltage (KVP)
0.020 and under	1,000	0.011 and under	8.5
.021 to .031	1,500	.012 to .015	10.0
.032 to .047	3,000	.016 to .020	11.3
.048 to .063	3,500	.021 to .040	12.5
.064 to .077	4,000	.040 to .062	13.8
.078 to .094	5,000	.063 to .074	15.0
		.075 and up	17.5

4.6.4 Dielectric withstanding voltage (see 3.4). Cables shall be tested as specified in method 6111 of FED-STD-228. The following exceptions shall apply:

a. The test shall be performed on finished cables only.
b. The source of power shall be not less than 5 kilovolt-amperes.
c. The crest factor of the test voltages shall not differ by more than ±10 percent from that of a sinusoidal wave when the transformer is loaded with the test specimen.
d. The immersion period shall be at least 6 hours for unshielded cables. Cables with an overall shield shall not be immersed in water, and shall be tested dry.
e. For unshielded cables, an alternating potential, as specified in table XIII, shall be applied between two terminals; one shall be each conductor in turn, and the other shall be all the remaining conductors and uninsulated ground wires, if present, tied together in electrical contact with the water.
f. For shielded cables, an alternating potential, as specified in table XIII, shall be applied between two terminals; one shall be each conductor in turn, and the other shall be all the remaining conductors, the uninsulated ground wires, if present, and the shield tied together. The test voltage shall be maintained for 60 seconds +5,-0 from the time it is reached.

4.6.4.1 Alternate dielectric withstanding voltage. The following alternations in procedure may be used when tested as specified in 4.6.4. Specimens are to be tested dry.

a. Arrange the conductors in a rectangle containing rows and columns; for example, a 20 conductor cable would have five columns of four rows; a 33 conductor cable would have five columns of seven rows, with two unused spaces.
b. Connect all the conductors in a column together. Do this for each column.
c. Apply the specified test voltage, for the specified time, between each column of the rectangle in turn, and the remaining columns connected together.
d. Disconnect all the conductors in a column. Do this for each column.
e. Connect all the conductors in a row together. Do this for each row.
f. Apply the specified test voltage, for 60 seconds +5, -0, between each row of the rectangle in turn, and the remaining rows connected together.
g. Connect all the conductors together.
h. If shields are present, apply the specified voltage from table XIII between all the conductors connected together and the shields connected together.

4.6.5 Insulation resistance (see 3.5). Insulation resistance of the cables shall be determined as specified in method 6031 of FED-STD-228, except that:

a. The test shall be performed on finished cables only.
b. The insulation resistance shall be measured immediately after the dielectric withstanding voltage test, and optionally after cable has been immersed at least six hours. The cable may be tested dry.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:40Z
Check the source to verify that this is the current version before use.

c. The test voltage shall be not less than 200 volts nor more than 500 volts dc.
d. The polarity of the conductor shall be maintained negative with respect to the water.
e. For unshielded cables, the potential shall be applied between two terminals, one being each conductor in turn, and the other being all the remaining conductors and uninsulated ground wires, if present, tied together in electrical contact with the water.
f. For shielded cables, the potential shall be applied between two terminals, one being each conductor in turn, and the other being all the remaining conductors, the uninsulated ground wires, if present, and the shields, tied together in electrical contact with the water. Cables with an overall shield shall not be immersed in water, and shall be tested dry.
g. If the measurement is made at a temperature other than 15.6°C, the manufacturer shall correct the measured value of insulation resistance to the resistance at 15.6°C. If the insulation resistance is equal to or greater than that required by 3.5, when the measurement is made at a temperature greater than 15.6°C, no correction factor need be used. The manufacturer shall demonstrate that the correction factor is accurate for his compound.
h. Insulation resistance may be determined in less than 1 minute, if the galvanometer has ceased fluctuating and the reading indicates that a steady insulation resistance value has been obtained.

4.6.6 Direct-current resistance (see 3.6). The direct-current resistance of each conductor and uninsulated ground wire, if present, shall be measured on the finished cables as specified in method 6021 of FED-STD-228, except the period of immersion shall be not less than 6 hours.

4.6.7 Surface resistance test (see 3.7). The insulated conductors and jacketed cables shall be tested in accordance with method 6041 of FED-STD-228. All specimens, after having been provided with the required electrodes but prior to testing, shall be cleaned by the procedure described in the test method. In positioning the specimens in the test chamber, the specimens shall be placed so that their ends are a minimum of 1 inch from any wall of the chamber.

4.6.8 Cold bend (see 3.8).

4.6.8.1 Specimens. One length of cable from each sample unit shall be prepared for test. The length shall be divided into two parts. Provide one specimen for checking inked markings and jacketed cable as a whole, and duplicate specimens for checking insulated conductors apart from the finished cable. The set shall be tested aged. If failure occurs to any one specimen, two additional specimens shall be prepared from the same sample unit and tested. If either of the additional specimens fail, the sample unit represented thereby shall be considered defective.

4.6.8.2 Procedure. The jacketed cable specimens and the insulated conductor specimens shall be conditioned in an air oven for 48 ±1/2 hours at 160° ±2°F and allowed to cool at room temperature. Aged specimens selected for checking inked markings and cables as a whole shall be attached to the proper size mandrel as determined in 4.6.8.2.1. The specimens selected for checking insulation apart from the cable shall have the jacket removed and one of each differently colored insulated conductor shall be attached to the proper size mandrel as determined in 4.6.8.2.1. The specimens shall be suspended vertically with lower ends weighted sufficiently to keep specimens taut and to permit bending them without handling. The mandrel and specimens shall be placed in the cold chamber at the lower table II temperature ±2°C for at least 20 hours. While at this temperature, the jacketed cable specimens shall be bent for six close turns, one turn for polyethylene insulation, around the mandrel at the rate of approximately 15 turns ±3 per minute; the insulated conductor specimens (shielding or nonadhering color coding removed) shall be bent at the rate of one turn per second. If six turns cannot be obtained from a specimen because its length is restricted by space limitations in the cold chamber, the amount of turns obtainable from a 5-foot specimen shall be permitted. After the test has been completed, the jacket on the specimens of cable shall be examined through a magnifying glass of at least 3-diameter magnification and then removed. The conductor insulation on all specimens shall be examined for cracks with the magnifying glass.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:40Z
Check the source to verify that this is the current version before use.

4.6.8.2.1 Mandrels. The mandrel shall be of a standard size in one of the diameters (in inches) specified in table XIV. The standard size selected shall be the largest size that does not exceed the value computed in accordance with table XV, except the 0.062-inch mandrel shall be used when a smaller size is indicated by the computation.

TABLE XIV. Mandrel diameters.

Inches	Inches	Inches	Inches
0.062	0.500	1.90	5.56
0.094	0.68	2.38	6.63
0.125	0.84	2.88	8.63
0.188	1.05	3.50	10.75
0.250	1.31	4.00	
0.375	1.66	4.50	

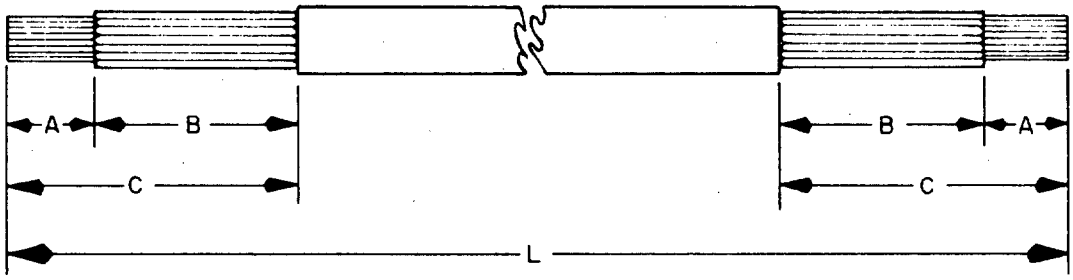
TABLE XV. Computed mandrel sizes.

Nominal outside diameter of insulated conductor or cable, inclusive 1/	Mandrel size, maximum		
	Unshielded cable	Shielded cable	Individual insulated conductor
Inch	Inches	Inches	Inches
Up to 0.150	1.0 X OD	3.0 X OD	1.0 X OD
.151 to .250	1.0 X OD	3.0 X OD	1.5 X OD
.251 to .300	1.0 X OD	3.0 X OD	2.0 X OD
.301 to .350	2.0 X OD	3.0 X OD	2.0 X OD
.351 to .450	2.5 X OD	3.0 X OD	2.5 X OD
.451 to .550	3.0 X OD	3.0 X OD	3.0 X OD
.551 to .750	4.0 X OD	4.0 X OD	
.751 to .950	5.0 X OD	5.0 X OD	
.951 to 1.500	6.0 X OD	6.0 X OD	
1.501 to 2.000	8.0 X OD	8.0 X OD	
2.001 and over	10.0 X OD	10.0 X OD	

1/ For nominal outside diameter of finished cable (see 1.2.1.7). The nominal outside diameter of insulated conductor shall be determined from the nominal outside diameter of the conductor and the nominal insulation thickness (see table V or VI). For parallel cables the nominal outside diameter shall be the major axis.

4.6.9 Bend (see 3.9). Samples of the completed cable prepared in accordance with figure 1 and connected in series (figure 2) shall be subjected to the tests in figure 3 at room temperature or 2000 cycles. A 110 or a 220-volt, 60 Hz ac potential with a prescribed load current of not less than 0.5 ampere shall be indicated on a recording ammeter to permit the determination of failure caused either by broken conductors or short circuits between conductors. Failure due to either cause shall terminate the test of a sample. The test procedures shown on figures 1, 2, and 3 are approved and a similar method shall be followed in making all mechanical tests. The value obtained shall be derived from test of three samples. If any of the samples tested do not comply with the requirements set forth, the lot shall be rejected.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:40Z
Check the source to verify that this is the current version before use.



A = 1.50
B = 2.00
C = 3.50

SET OF SAMPLES TO CONSIST OF

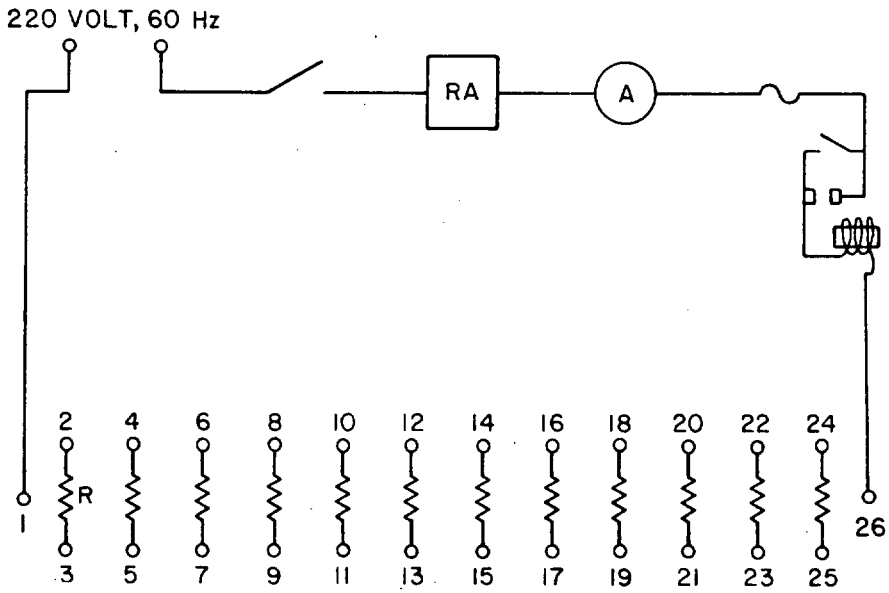
6 SAMPLES FOR IMPACT, L = 16.00
3 SAMPLES FOR BEND, L = 42.00
3 SAMPLES FOR TWIST, L = 66.00

INCHES	MM
1.50	38.1
2.00	50.8
3.50	88.9
16.00	406.4
42.00	1066.8
66.00	1676.4

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.

FIGURE 1. Cable samples.

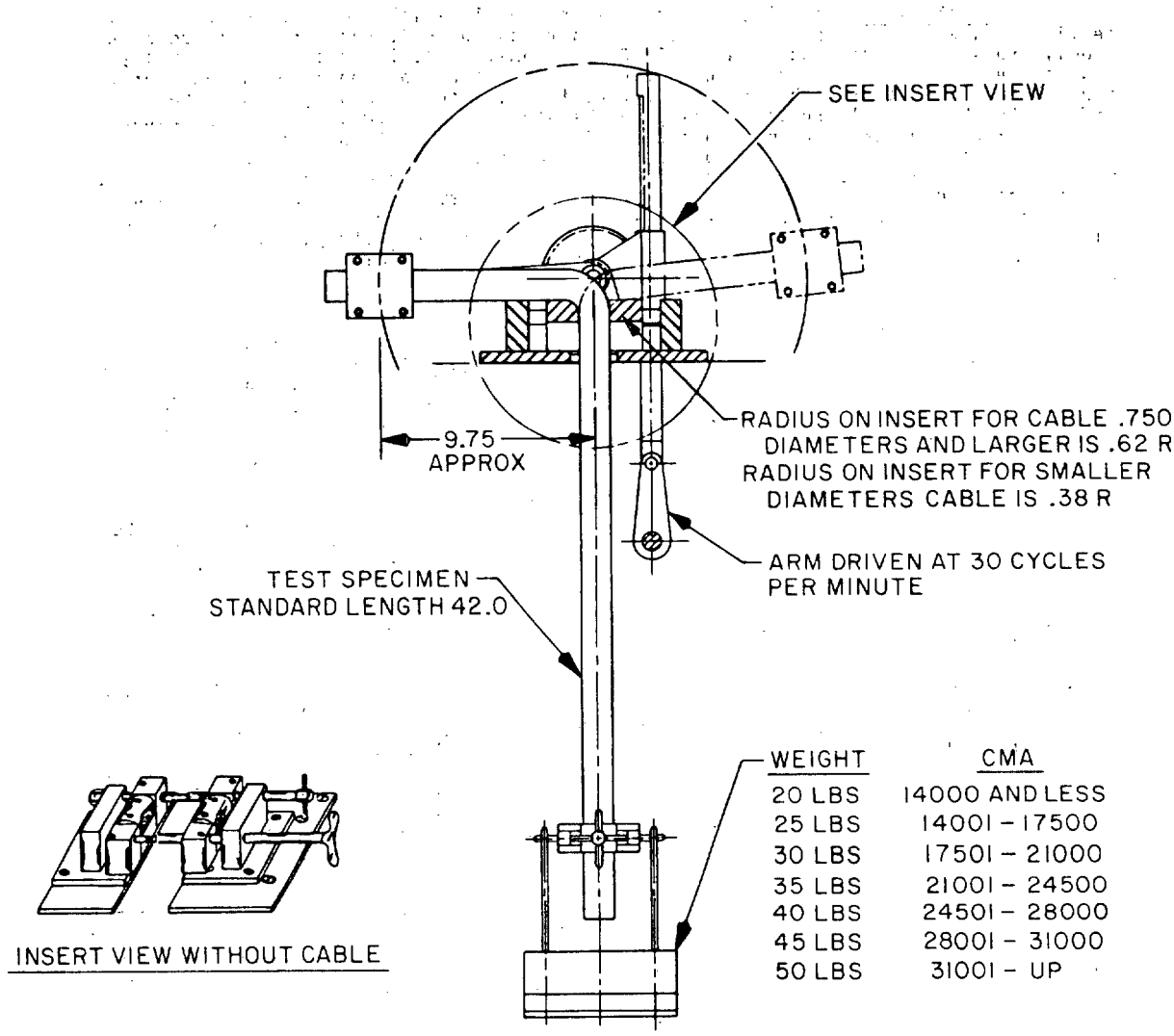
Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:40Z
Check the source to verify that this is the current version before use.



R = 20 OHM RESISTOR
A = AMMETER
RA = RECORDING AMMETER

FIGURE 2. Fixture, cable testing electrical circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:40Z
Check the source to verify that this is the current version before use.



WEIGHT	CMA
20 LBS	14000 AND LESS
25 LBS	14001 – 17500
30 LBS	17501 – 21000
35 LBS	21001 – 24500
40 LBS	24501 – 28000
45 LBS	28001 – 31000
50 LBS	31001 – UP

INCHES	MM
.38	9.7
.62	15.7
.750	19.05
9.75	247.6
42.0	1067

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. During test a 60 hertz current of not less than 0.5 amps flow through conductors.
4. 1 cycle shall consist of 90° bend each side and return to original position.

FIGURE 3. Bend test, fixture, cable testing.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:40Z
Check the source to verify that this is the current version before use.

4.6.10 Durability (see 3.10). A short specimen of finished wire or sheath shall be firmly clamped in a horizontal position with its upper longitudinal surface area freely exposed. A small steel mandrel (0.025 diameter ±0.001) shall be repeatedly rubbed over the insulation surface at the stripe so that the longitudinal axis of the mandrel and specimen will be at right angles to each other. A weight shall be affixed to the jig holding the rubbing mandrel so that the combined jig and weight exerts a 500-gram thrust normal to the insulation surface. A motor-driven reciprocating cam mechanism and counter shall be used to permit an accurately measured number of abrasion strokes. The length of the stroke in one direction shall be 3/8 inch and the frequency of the stroke shall be 120 strokes per minute (each stroke consisting of a 180 degree-rotation of the eccentric drive mechanism). The direction of motion shall be along the axis of the wire or cable jacket and perpendicular to the axis of the mandrel. This procedure shall be repeated on one additional specimen selected from at least 50 feet apart.

4.6.11 Polyethylene shrink-back (see 3.11). A 7-inch long specimen shall be cut from the insulated conductor. Shield and color coding braid, tape, or jacket shall be removed. The specimen shall then be cut to 6 inches with conductor flush with the insulation. The specimen shall then be air-oven aged at 85° ±1°C for 24 hours ±1/2. The specimen shall then be removed and allowed to cool to room temperature. The amount the conductor extends beyond the insulation shall be measured at each end, and the insulation shall be examined for splitting or cracking.

4.6.12 Polyamide heat stability (see 3.12).

4.6.12.1 Polyamide coatings over primary insulation. A 1-foot specimen shall be bent around a metal mandrel six times the outside diameter of the specimen for two turns and taped down on its ends. The specimen and the mandrel shall be placed in a gravity convection type oven at a temperature of 200°F ±5° for 24 hours. Remove specimen and mandrel from oven and cool in a silica gel desiccator or equal until it returns to room temperature (1 hour minimum). Remove from desiccator and straighten the specimen.

4.6.12.2 Polyamide coating under or over component shield. A sufficient length of specimen shall be used to insure at least six close turns when bent on a metal mandrel, six times the outside diameter of the specimen. Each turn shall be in contact with the next. The ends of the specimen shall be taped together so that the turns are held in place on the mandrel. The mandrel and specimen shall be suspended in a gravity convection type oven at a temperature of 300°F ±5° for 15 minutes. The mandrel and specimen shall be removed from the oven and cooled in a silica gel desiccator or equal until the specimen is at room temperature (1 hour minimum). Remove from the desiccator and unwrap the coils.

4.6.13 Fluid immersion. Four individual samples approximtely 24 inches in length shall be stripped to the bare conductor on either end for 1 inch and formed in a loop not less than 5 times the diameter of the cable or wire. A minimmum of 18 inches of the cable or wire shall then be placed in each of the fluids specified in table XVI for the time specified. Upon completion of the soaking and air drying, the increase in diameter shall be measured.

TABLE XVI. Fluids.

Fluid	Soak time (hours	Air-drying time (hours)
Hydrocarbon fluid (Specification TT-S-735, type III)	4	2
Isopropyl alcohol (Specification TT-I-735)	4	2
Carbon tetrachloride (Specification Q-C-265)	4	2
Kerosene, 50 percent aircraft lubricating oil, 50 percent at 48° to 50°C (Specifications ASTMD 3699 and MIL-L-6082, grade 1100, respectively)	16	3

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:40Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Packaging requirements. Except as specified herein, the requirements for packaging shall be in accordance with MIL-C-12000. The type designation (see 1.2.1) shall be included in the identification marking of all interior and exterior containers.

6. NOTES

6.1 Intended use. The cables covered by this specification are intended for use by the Armed Services in extensive electrical and electronic applications. The cables may be used to transmit power, synch pulses, data transmission voltages, broadband-, audio-, or control-power. The shielded cables are suitable for radio-frequency use in limited applications. A tough and flexible jacket is essential because normally the cables will be subjected to extreme mechanical abuse and extreme humidity and temperature conditions. Cables designated for light-, medium-, and heavy-duty are for use as indicated in 6.1.1 through 6.1.3. Commercial type cables similar in construction to the cables described herein are indicated in 6.1.4. These commercial types should be replaced with MIL-C-3432 cable constructions.

6.1.1 Light-duty (L) cables. Light-duty cables are intended for use in test equipment in short lengths, or for interconnection of major components. They are intended to withstand severe flexing and frequent manipulation. Light-duty cables should not be used where they will be stepped on, run over by vehicles, beaten, or subjected to severe impacts. Light-duty cables are suitable for lightweight portable tools or small motor and generator leads where flexibility rather than long life is essential.

6.1.2 Medium-duty (M) cables. Medium-duty cables are intended to withstand the same usage as heavy-duty cables except they should not be used where they will be run over by vehicles or be subjected to severe impacts. They are intended to be a substitute for all uses of heavy-duty cables when the reduction in weight would be advantageous to the equipment they are used in. Medium-duty cables are suitable for small portable tools, sound equipment, radio receivers, and motor leads which do not require the heavier, sturdier, heavy-duty cables.

6.1.3 Heavy-duty (H) cables. Heavy-duty cables are intended for use where they will be subjected to extreme service impacts or will be run over by heavy vehicles, such as trucks, tanks, or the like. They are designed to withstand severe flexing and mechanical abuse over long periods of time without deterioration. Heavy-duty cables are suitable for portable tools, extension lamps, charging cables, and control cables.

6.1.4 Similar commercial types of cables. Commercial types of cables similar in construction and use to cables covered by this specification are as follows:

Light-duty (L) - - - - - - - - - - commercial type SV.
Medium-duty (M)- - - - - - - - - - commercial type SJ.
Heavy-duty (H) - - - - - - - - - - commercial type S.

6.1.5 Temperature applications. Cables designated for medium low temperature applications (G, O, E, T) are intended for use from -40° to +85°C; (L, D, M, N) for -55° to 85°C; (H) from -55° to +90°C; (B, C, and K) from -55° to 80°C.

6.1.6 Length of lay. For extra-flexible and flexible cable, the length of lay should nominally be 10 to 12 times the layer outer diameter. When all elements are of the same size, the lay may be as high as the upper limits (see 3.2.6.1). When the cable is a mixture of large and small elements, the length of lay should be closer to 8 in extra-flexible cable.

6.1.7 Reinforcement. Nylon yarn, 840/1 denier, suitably treated (phenol-latex coated) for bonding to rubber during cure is recommended (see 3.2.7).

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Complete type designation (see 1.2.1).
c. Lengths required.

6.3 Supersession data. This specification supersedes MIL-C-3432D, dated 5 July 1966.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:40Z
Check the source to verify that this is the current version before use.

6.3.1 MIL-C-5756 supersedure. Cables similar and equal to those of MIL-C-5756 may be specified under MIL-C-3432 as follows:

Conductor size	MIL-C-3432 Class	Flexibility
18 - 16	HN	F (flexible)
14 - 10	HN	E (extra-flexible)
8 - 4/0	HN	M (max-flexible)

Example: CO-03HNM (3/0000) - Represents a MIL-C-3432 type low temperature, oil resistant, heat resistant, heavy duty cable with three maximum flexibility size 4/0 conductors insulated in natural rubber or a blend of natural rubber and synthetic polyisoprene and jacketed over-all with chloroprene rubber. This corresponds to MIL-C-5756, three conductors, size 4/0. The color code for 3 and 5 conductor of MIL-C-3432 cables are different from the color code of MIL-C-5756, but complies with national and military safety standards.

6.4 Definitions.

6.4.1 Conductor. A conductor consists of a solid or stranded wire, with insulating covering, suitable for carrying an electric current.

6.4.2 Equipment ground wire. An insulated conductor (green) intended for grounding non-current carrying metal parts of equipment.

6.4.3 Neutral conductor (cables with three or more conductors). The conductor that is intended to be so energized that, in the normal steady state, the voltages from every other conductor to the neutral conductor, at the terminals of entry of the circuit into a delimited region, are definitely related and usually equal in amplitude.

NOTE: If the circuit is an alternating-current circuit; it is intended also that the voltage have the same period and the phase difference between any two successive voltages, from each of the other conductors to the neutral conductor, selected in a prescribed order, having a predetermined valve usually equal to 2 radians divided by the number of phase conductors. (Extracted from " IEEE Standard Dictionary of Electrical and Electronic Terms".)

6.4.4 Wire. A single metallic conductor of stranded construction designed to carry current in an electrical circuit.

6.4.5 Uninsulated ground wires. Uninsulated ground wires are wires bare or covered with a green covering to identify them as ground wires and should not be confused with the insulated green conductor used in portable power cable which is used as the equipment ground conductor (see 6.4.2). These insulated ground wires are used in heavy duty outdoor and mine cables.

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - CR
Navy - YD
Air Force - 85

Review activities:
Army - AR, MI
Navy - AS, SH
Air Force - 17, 99
NSA -
DLA - IS

User activities:
Navy - MC
Air Force - 11, 15

Preparing activity:
Army - CR

Agent:
DLA - ES

(Project 6145-0750)

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:40Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.

NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-C-3432E Cables, (Power and Special Purpose and Wire, Electrical)(300 and 6000 Volts)

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

Headquarter,
US ARMY CECOM
ATTN: DRSEL-PED-CM-DM
Ft Monmouth, NJ 07703

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*	TELEPHONE NO.
	DATE

DD FORM 1 OCT 76 1426 EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:40Z
Check the source to verify that this is the current version before use.